Suggestions for final project papers:

Sample covariance and Wigner-type matrices:

- Bai (1997) “Circular law”
- Ben Arous, Guionnet (1997) “Large deviations for Wigner’s law and Voiculescu’s non-commutative entropy”
- Bai, Silverstein (1998) “No eigenvalues outside the support of the limiting spectral distribution of large-dimensional sample covariance matrices”
- Bai, Silverstein (2004) “CLT for linear spectral statistics of large-dimensional sample covariance matrices”
- Erdos, Knowles, Yau, Yin (2013) “The local semicircle law for a general class of random matrices”
- Srivastava, Vershynin (2013) “Covariance estimation for distributions with 2+epsilon moments”
- Chafai, Tikhomirov (2018) “On the convergence of the extremal eigenvalues of empirical covariance matrices with dependence”
- Erdos (2025) “Lecture notes on zigzag strategy”

Random matrices and free probability theory:

- Haagerup, Thorbjornsen (2005) “A new application of random matrices: Ext(C_red*(F_2)) is not a group”
- Benaych-Georges (2009) “Rectangular random matrices, related convolution”
- Benaych-Georges, Nadakuditi (2011) “The eigenvalues and eigenvectors of finite, low rank perturbations of large random matrices” / “The singular values and vectors of low rank perturbations of large rectangular random matrices”
- Mingo, Speicher (2012) “Sharp bounds for sums associated to graphs of matrices” / Chapter 4.4 of *Free Probability and Random Matrices*
- Au, Cebron, Dahlqvist, Gabriel, Male (2021) “Freeness over the diagonal for large random matrices”
- Bandeira, Boedihardjo, van Handel (2023) “Matrix concentration inequalities and free probability”

Non-linear random matrices / applications:

- Johnstone (2001) “On the distribution of the largest eigenvalue in Principal Components Analysis”
- El Karoui (2010) “The spectrum of kernel random matrices”
- Ma (2013) “Sparse Principal Component Analysis and iterative thresholding”
- Auffinger, Ben Arous, Cerny (2013) “Random matrices and complexity of spin glasses”
- Pennington, Worah (2017) “Nonlinear random matrix theory for deep learning”

- Bordenave, Lelarge, Massoulie (2018) “Nonbacktracking spectrum of random graphs: Community detection and non-regular Ramanujan graphs”
- Fan, Wang (2020) “Spectra of the conjugate kernel and neural tangent kernel for linear-width neural networks”
- Hastie, Montanari, Rosset, Tibshirani (2022) “Surprises in high-dimensional ridgeless least squares interpolation”
- Lu, Yau (2025) “An equivalence principle for the spectrum of random inner-product kernel matrices with polynomial scalings”

You are welcome to choose a paper not listed here, with permission of the instructor. If your paper has already been selected, you will be asked to choose a different one.